

# Data Science

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## Lecture 06: Decision Trees and Ensembles

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# Module 1

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## Trees as Hypotheses

**Guiding Question:** *A linear model hides its reasoning in 200 weights. What would a model look like that can be read aloud?*

# A Tree Is a Sequence of Questions

Each inner node tests one attribute, each branch is an answer, each leaf is a class.

| Example | Sky   | Temperature | Humidity | Wind   | EnjoySport |
|---------|-------|-------------|----------|--------|------------|
| 1       | sunny | warm        | normal   | strong | yes        |
| 2       | sunny | warm        | high     | strong | yes        |
| 3       | rainy | cold        | high     | strong | <b>no</b>  |
| 4       | sunny | warm        | high     | strong | yes        |

To classify a new day, start at the root and follow the answers to a leaf.

The tree **is** the hypothesis. Unlike a weight vector, it can be read aloud as a set of rules — which is why trees are the standard choice when a decision has to be explained.

# Construction and Classification

## Construction (*recursive*)

If the data is pure enough, make a leaf.  
Otherwise: choose a **split criterion**,  
decompose  $D$  into  $D_1, \dots, D_s$ , and recurse  
on each part.

## Classification (*recursive*)

If the node is a leaf, return its class.  
Otherwise: follow the successor selected  
by the example's attribute value.

The whole learning problem reduces to one question: **which attribute to split on?**  
Everything else is bookkeeping.

# When to Use Them

## Well suited

Attribute-value data; discrete or continuous features; missing values; a need for an explainable model; no assumption of linearity.

## Poorly suited

Smooth relationships (a tree approximates a diagonal boundary with a staircase); many equally relevant features; and — as we will see — stability.

# Mini-Exercise

Using the four examples above, write down **by hand** a decision tree that classifies all of them correctly.

1. Can you do it with a single test?
2. How many different correct trees can you find?
3. Which of them would you trust on a new day, and why?

# Mini-Exercise — Solution

# Mini-Exercise — Solution

**1.** Yes — splitting on Sky alone separates them: sunny → yes, rainy → no.

**2.** Several. Temperature (warm/cold) also separates them perfectly, as do combinations that test both.

**3.** Undecidable from four examples. Perfect training accuracy does not choose between competing hypotheses — this is the inductive-bias problem, and the next module gives it a criterion.

# Takeaway

A decision tree is a **readable hypothesis** built by recursive splitting. The only real decision is which attribute to split on next — and the data alone does not settle it.

# Module 2

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## Choosing a Split

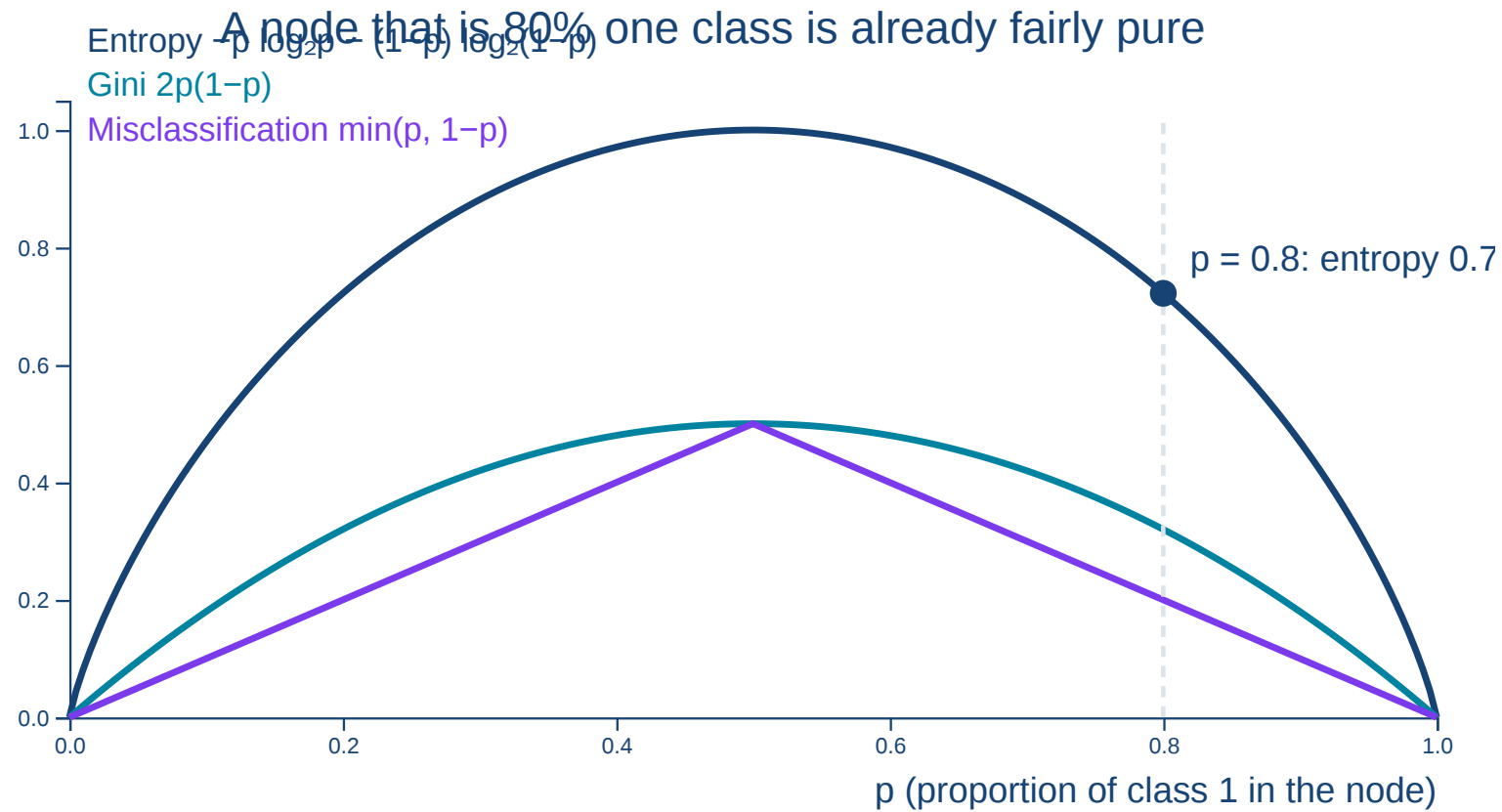
**Guiding Question:** *Every attribute could be split on. What makes one split better than another — and how do we measure it?*

# What Makes a Split Good?

A split is good if the resulting subsets are **purier** than what we started with — closer to containing a single class.

So we need a number that measures impurity: maximal when the classes are evenly mixed, zero when a node is pure.

# Three Measures



# Entropy

For a node with class probabilities  $P(A_1), \dots, P(A_k)$ :

$$H(\mathcal{A}) = - \sum_{i=1}^k P(A_i) \log_2 P(A_i)$$

measured in **bits** — the average number of yes/no questions needed to identify the class of an example in that node.

## Pure node

$H = 0$  — no question needed

## Two classes, 50/50

$H = 1$  bit — one question needed

## $k$ equal classes

$H = \log_2 k$  bits

# Conditional Entropy and Information Gain

After splitting on an attribute  $\mathcal{B}$  with outcomes  $B_1, \dots, B_s$ :

$$H(\mathcal{A} \mid \mathcal{B}) = \sum_{j=1}^s P(B_j) \cdot H(\mathcal{A} \mid B_j)$$

— the impurity of each part, weighted by how much data lands in it.

$$\text{Gain}(\mathcal{B}) = H(\mathcal{A}) - H(\mathcal{A} \mid \mathcal{B})$$

**Information gain** is the impurity removed by the split. Choose the attribute that maximises it.

# Mini-Exercise

A node holds 8 examples: 4 "yes", 4 "no". An attribute splits it into [3 yes, 1 no] and [1 yes, 3 no].

1. What is the entropy before the split?
2. What is the conditional entropy after it?
3. What is the information gain?

# Mini-Exercise — Solution

# Mini-Exercise — Solution

**1.**  $H = 1$  bit — a perfectly even mix.

**2.** Each part has  $H = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} \approx 0.811$ ; both hold half the data, so  $H(\mathcal{A} \mid \mathcal{B}) \approx 0.811$ .

**3.** Gain  $\approx 1 - 0.811 = 0.189$  bits.

A split that leaves both branches 3:1 removes less than a fifth of a bit.

# Takeaway

Impurity measures how mixed a node is; **information gain is the impurity a split removes**, weighted by the data it moves. Entropy and Gini agree closely; misclassification error is too flat to choose between splits.

# Module 3

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## Building the Tree

*Guiding Question: ID3 never reconsiders a split. What does that greedy choice assume about the world?*

# ID3

**ID3**(examples  $D$ , attributes  $A$ , target)

1. If all examples share one class  $\rightarrow$  return a leaf with that class.
2. If  $A$  is empty  $\rightarrow$  return a leaf with the majority class.
3. Otherwise choose  $A^* \in A$  with the highest **information gain**.
4. For each value  $a$  of  $A^*$ : add the subtree **ID3**( $D_a$ ,  $A \setminus \{A^*\}$ , target).

Greedy and recursive: each split is chosen for immediate gain, and never reconsidered.

# What ID3 Assumes

## Hypothesis space

**All** decision trees over the attributes — so the target function is certainly in there. No representational bias.

## Search bias

It prefers trees that place **high-gain attributes near the root**, and therefore **shorter trees**. It stops at the first tree that fits.

Is a preference for short trees justified? It is Occam's razor as an algorithm — a genuine assumption about the world, not a mathematical necessity.

# CART

## What differs

**Binary** splits only **Gini** impurity instead of entropy handles **regression** by splitting on variance reduction and predicting the mean in each leaf.

## What does not

The template is unchanged: choose a split, decompose, recurse. CART is a different set of choices inside the same skeleton.

Binary splits avoid a defect of multi-way splits: an attribute with many values (an ID number, in the extreme) achieves high information gain by fragmenting the data into pure singletons that generalise to nothing.

# Mini-Exercise

Your dataset includes a `customer_id` column, unique for every row.

1. What information gain does splitting on it achieve?
2. What does the resulting tree look like?
3. What does it predict for a new customer?

# Mini-Exercise — Solution

# Mini-Exercise — Solution

**1.** Maximal — every branch holds one example, so every leaf is pure and the conditional entropy is 0.

**2.** A root with one branch per training example: a lookup table, one level deep.

**3.** Nothing useful — a new customer's id matches no branch. Perfect training purity, zero generalisation.

This is why gain must be corrected for the number of values (gain ratio), or splits restricted to binary (CART).

# Takeaway

ID3 is a **greedy, non-backtracking search** whose bias is a preference for short trees. CART changes the split type and the impurity measure, not the idea. Both can be fooled by an attribute with too many values.

# Module 4

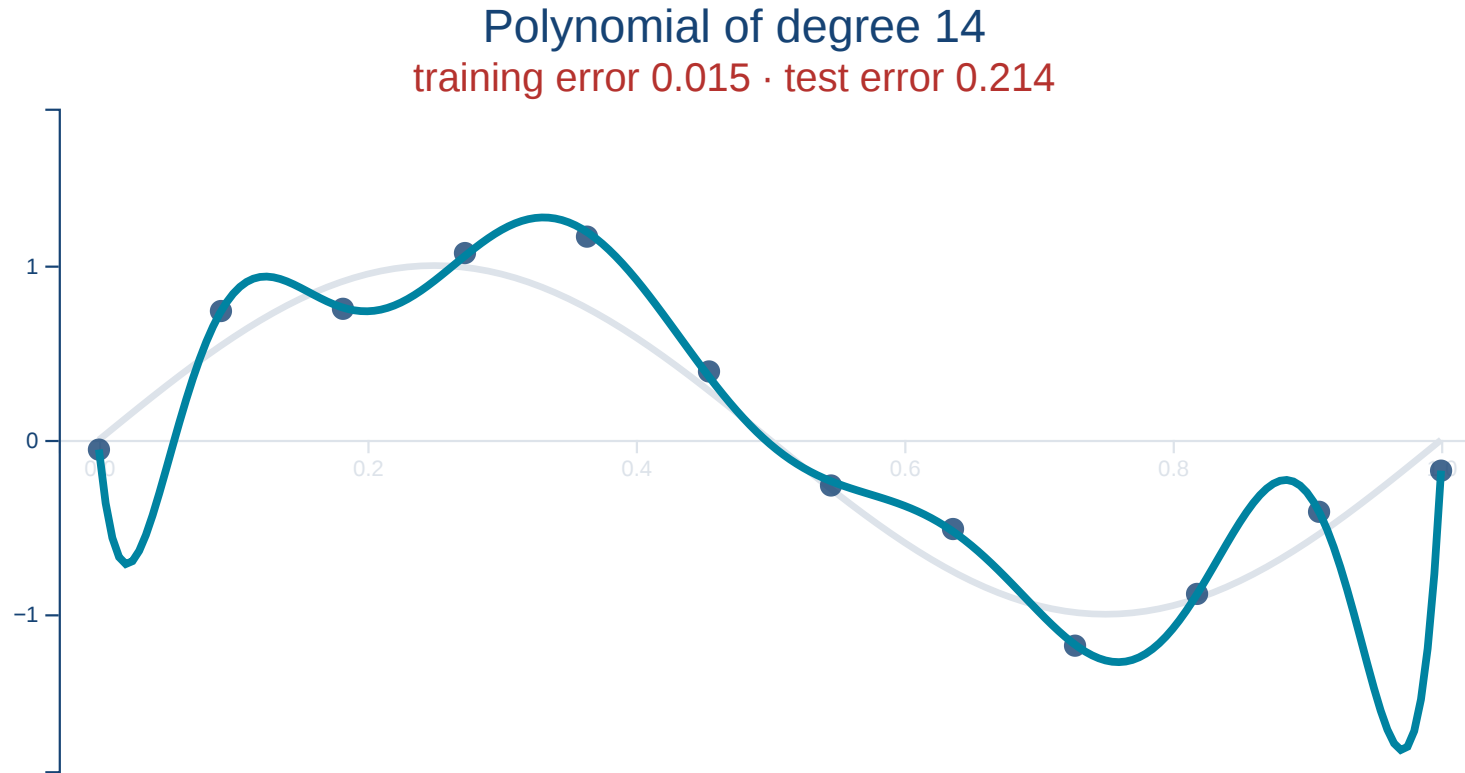
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## Overfitting and Pruning

**Guiding Question:** *A tree can achieve 100% training accuracy on any dataset. Why is that a symptom rather than an achievement?*

# A Tree Grown to Purity Is Overfitted

Left alone, ID3 splits until every leaf is pure — memorising the training set, noise included.



Tree depth plays the role of polynomial degree: more splits, lower training error, and eventually rising test error.

# Two Places to Intervene

## Stopping (pre-pruning)

Halt splitting **during** construction — when a node is small, when gain is below a threshold, or at a depth limit.

Cheap, but short-sighted: a weak split may enable a strong one beneath it.

## Pruning (post-pruning)

Grow the full tree, then **remove** subtrees that do not earn their keep.

More expensive, and generally more reliable.

# Reduced Error Pruning

1. Split the data into a **training** and a **validation** set.
2. Grow the full tree on the training set.
3. For each inner node: consider replacing its subtree with a leaf carrying the majority class.
4. Keep the replacement if accuracy on the **validation** set does not fall.
5. Repeat until no beneficial replacement remains.

The decision is made on data the tree was **not** grown on. Pruning against the training set would remove nothing — every subtree helps there by construction.

# Mini-Exercise

A tree has 100% training accuracy with 340 leaves. After pruning it has 41 leaves, 94% training accuracy and 89% validation accuracy — up from 78%.

1. What did the 299 removed leaves contain?
2. Why did training accuracy fall?
3. Was pruning worth it?

# Mini-Exercise — Solution

# Mini-Exercise — Solution

**1.** Mostly noise — splits that separated individual training examples without capturing structure that recurs.

**2.** Because those examples are now misclassified. That is the intended outcome, not a regression.

**3.** Yes: +11 points on unseen data, and a tree small enough to read. Both axes improved.

# Takeaway

An unpruned tree memorises. **Pruning trades training accuracy for generalisation**, and the trade must be judged on data the tree never saw.

# Module 5

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## Ensembles

**Guiding Question:** *Change a few examples and the tree changes completely. Can that instability be turned into an advantage?*

# Unstable Learners

## High variance

Many parameters, so small changes in the training data produce very different models.

## Runtime limits

The learner cannot search its hypothesis space exhaustively and settles for a heuristic path.

## Insufficient capacity

The model family is too simple to express the target at all.

**Decision trees are the textbook unstable learner:** change a handful of examples and a different attribute wins at the root, producing a completely different tree.

# The Idea

**Ensemble learning** builds several learners and combines their predictions — by majority vote for classification, by averaging for regression.

## Bagging

**Bootstrap aggregating** — train on resampled datasets.

## Boosting

Train weak learners in sequence, each focusing on the previous one's mistakes.

## Random forest

Bagging plus randomised attribute selection at every split.

Ensembles exploit instability: **many different wrong models, averaged, are right more often than any one of them.**

# Bootstrap Aggregating

**The problem.** A group of predictors helps only if they differ — but we have one algorithm and one dataset, which yield one model.

**The solution.** Manufacture different datasets: draw  $n$  examples from  $D$  **with replacement**, train a predictor on each sample, and combine them by vote or average.

The learner never changes. Only the data it sees does.

# What Resampling Actually Does

Each bootstrap sample contains about **63.2%** of the distinct examples of  $D$ . The probability that a particular example is *never* drawn in  $n$  tries is

$$\left(1 - \frac{1}{n}\right)^n \xrightarrow{n \rightarrow \infty} e^{-1} \approx 0.368$$

## The gain

Each model sees a different two-thirds of the data, so their errors are partly independent. **Averaging reduces variance** — and therefore overfitting.

## The left-out third

The ~36.8% not drawn is the *out-of-bag* sample, usable as a free validation set for that model.

# Mini-Exercise

You bag 100 trees. Each individually achieves 70% accuracy.

1. Under what condition would the ensemble beat 70%?
2. What happens if all 100 trees are identical?
3. Does bagging reduce bias, variance, or both?

# Mini-Exercise — Solution

# Mini-Exercise — Solution

**1. Their errors must be at least partly independent** — different trees must be wrong about different examples for the vote to correct them.

**2. Nothing improves:** 100 copies of one model vote unanimously and reproduce its 70%.

**3. Mainly **variance**.** Averaging does not fix a systematic error every model shares.

# Takeaway

Bagging manufactures diversity by **resampling the data**, and averages away variance. It helps exactly to the extent that the models' errors are independent — and not at all when they are not.

# Why Bagging Trees Is Not Enough

## Trees stay similar

If one attribute has much higher gain than the rest, it wins at the root of **every** bootstrap sample. The trees correlate, and averaging them buys little.

## And it costs

Interpretability is gone — you can no longer read the model as a set of rules — while the computation of 100 near-identical trees is largely wasted.

# The Fix: Randomise the Split Too

## Random forest (Breiman 2001)

For each of  $T$  trees:

1. Draw a bootstrap sample from  $D$ .
2. Grow a tree — but **at every node**, choose the split from a random subset  $A' \subset A$  of  $m_A$  attributes rather than from all of them.
3. Do not prune.

Combine by majority vote (classification) or average (regression).

The dominant attribute is simply **absent** from most nodes' candidate sets, so other attributes get to define the tree. Diversity is engineered rather than hoped for.

# What You Get and What You Give Up

## Gained

Strong accuracy with almost no tuning out-of-bag error for free a usable attribute-importance measure.

## Lost

The readable single tree. A forest is a **committee**, and no member speaks for it.

Attribute importance from a forest is not a causal statement. It reports which attributes the trees found useful for splitting — under this data, this sampling, this correlation structure.

# Mini-Exercise

You must choose between a single pruned decision tree (86% accuracy) and a random forest (91%) for:

1. an internal churn-prediction model,
2. an automated loan-refusal system that must be justified to applicants.

Which do you choose in each case?

# Mini-Exercise — Solution

# Mini-Exercise — Solution

1. The **forest**: five accuracy points matter and nobody needs the model explained.

2. The **tree** — or a forest plus a documented, auditable process. A refusal that cannot be explained may be neither acceptable nor lawful.

The best model is the one that fits the **decision**, not the leaderboard.

# Takeaway

Random forests **engineer diversity** by randomising the attributes available at each split, turning unstable trees into a stable committee — at the cost of the one thing a single tree was best at: being readable.